

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Mixed Factoring Practice

Example (Factor completely):

$2x^2 - 8$ — GCF is 2: $2(x^2 - 4)$. Then $x^2 - 4$ is a difference of squares: $2(x + 2)(x - 2)$.

Directions: Factor completely. Always check for a GCF first.

1. $3x^2 - 27$

4. $x^2 - 64$

2. $x^2 + 9x + 20$

5. $x^2 - 3x - 10$

3. $5x^2 + 10x$

Challenge Problems

6. $4x^2 - 36$ — factor completely (GCF then a pattern).

Answer Key and Procedure

Procedure

1. Factor out the GCF.
2. Count the terms: two squares $\rightarrow$ difference of squares; three terms $\rightarrow$ product-sum.
3. Factor the leftover and check by distributing.

Answers

1. $3(x + 3)(x - 3)$
2. $(x + 4)(x + 5)$
3. $5x(x + 2)$
4. $(x + 8)(x - 8)$
5. $(x - 5)(x + 2)$
6. $4(x + 3)(x - 3)$